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KARNATAKA

MINI-PROJECT REPORT

ON

"AI MINI CHATBOT"

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE ARTIFICIAL INTELLIGENC(BCS515B) COURSE OF V SEMETER

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CHAPTER 1

INTRODUCTION TO AI TOOLS

Artificial Intelligence (AI) is one of the most rapidly growing technologies in the world today. It refers to the ability of machines and computer systems to perform tasks that normally require human intelligence. Some of these tasks include understanding natural language, recognizing images, making decisions, learning from data, and solving problems. AI systems can analyze huge amounts of information, identify patterns, and perform actions without needing human intervention.

AI tools are software applications designed to implement AI concepts efficiently. These tools make it easier to build intelligent systems without manually writing complex algorithms. Examples include chatbots for communication, virtual assistants like Siri and Alexa, image recognition software, recommendation systems (like YouTube and Netflix), and automated data analysis tools used in industries.

AI tools rely on various branches of AI, such as Machine Learning (ML), Deep Learning (DL), and Natural Language Processing (NLP). Machine Learning focuses on training models using data. Deep Learning uses neural networks that function similarly to the human brain. NLP helps machines understand human language and generate meaningful responses.

This project focuses on creating a **Mini AI Chatbot** using basic web technologies like HTML, CSS, and JavaScript. While the chatbot does not use complex machine learning methods, it explains how AI-based question–answer systems function at a basic level. The chatbot can answer predefined questions related to Artificial Intelligence. The project helps students understand fundamental AI concepts, how chatbots work, and how rule-based systems store and retrieve information.

CHAPTER 2

PROBLEM STATEMENT

Modern chatbots like ChatGPT, Google Bard, and Alexa use advanced machine learning and NLP techniques. However, for beginners, understanding the inner working of such systems can be challenging. Therefore, the objective of this project is to design a simple, rule-based AI Mini Chatbot that can answer common AI-related questions.

The problem statement can be defined as:

"To develop a mini chatbot system using basic web programming that can answer predefined AI questions and demonstrate the fundamental working of conversational AI."

The project aims to create a chatbot that:

- Accepts user input through a text box
- Matches the input with predefined questions
- Gives the correct response stored internally
- Displays a default answer for unknown questions
- Shows a simple and attractive interface
- Teaches beginners how chatbots work internally

The system must be simple, lightweight, and beginner-friendly. It should not require external libraries, APIs, or advanced algorithms. Instead, it should rely purely on JavaScript-based string matching.

This small-scale chatbot will help users understand:

- How AI responses are generated
- How rule-based systems function
- How user queries are processed
- How UI and logic connect in a web-based project

Thus, the project solves the problem of learning chatbot fundamentals in the easiest way.

CHAPTER 3

WORKING PROCEDURE

The entire working process of the AI Mini Chatbot can be explained in the following steps:

1. **Identify the Questions**

   Five commonly asked AI questions are selected:

   - What is AI?
   - What is Machine Learning?
   - What is Deep Learning?
   - What is NLP?
   - What is a Chatbot?

2. **Store Questions & Answers**

   Each question is stored as a key in a JavaScript dictionary (object), and each answer is linked to its corresponding key.

3. **User Input**

   The user types a question in the input field.

   Example: *"What is AI?"*

4. **String Matching**

   The input is converted to lowercase and matched with the stored questions.

5. **Response Generation**

   - If the question matches → Show the predefined answer.
   - If no match → Show a default message:

     "Sorry, I know only basic AI questions."

6. **Display Output**

   The generated response appears inside the chatbot box.

7. **User Interface**

   A simple UI is created using HTML and CSS with a heading, decent background color, and footer copyright.

This procedure demonstrates the basic working of rule-based chatbots.

CHAPTER 4

WORKING CODE

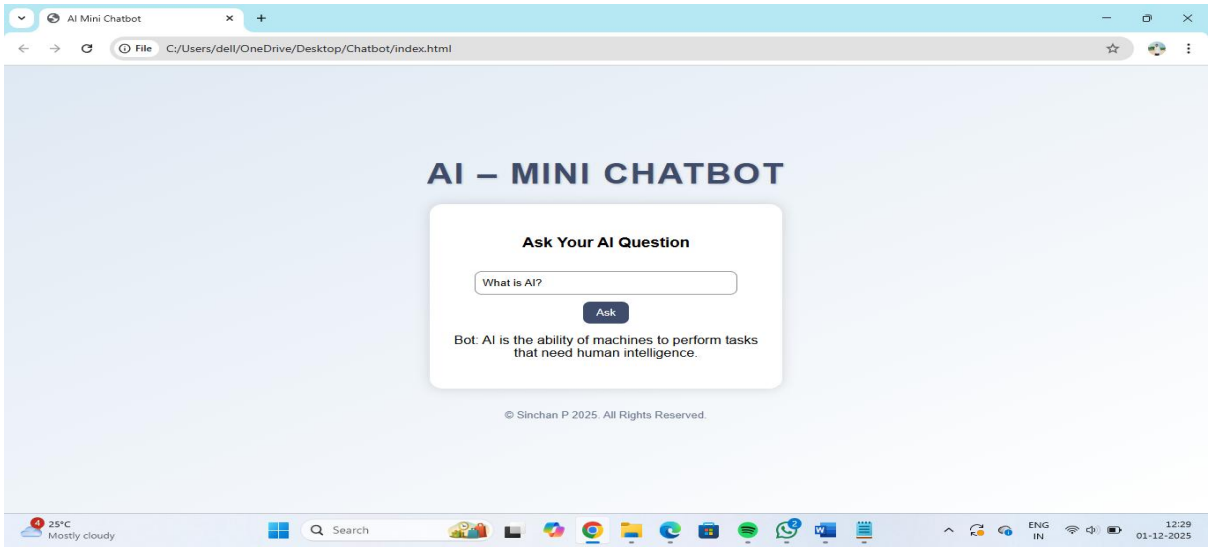
Below is the complete working code for the AI Mini Chatbot (HTML + CSS + JavaScript):

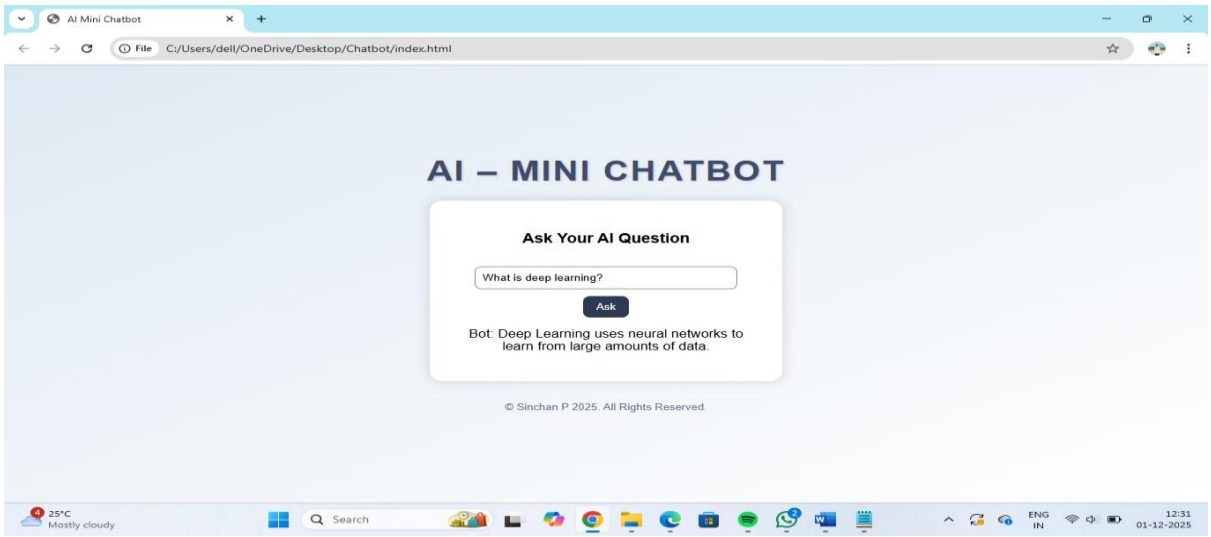
```
<!DOCTYPE html>
<html>
<head>
  <title>AI-MINI CHATBOT</title>
  <style>
    body {
      background: #e8f0fe;
      font-family: Arial;
      text-align: center;
    }
    #chatbox {
      width: 40%;
      margin: auto;
      padding: 20px;
      background: white;
      border-radius: 15px;
      box-shadow: 0 0 10px gray;
    }
    input { width: 80%; padding: 10px; margin-top: 20px; }
    button { padding: 10px 20px; cursor: pointer; }
    footer { margin-top: 30px; font-size: 13px; color: #555; }
```

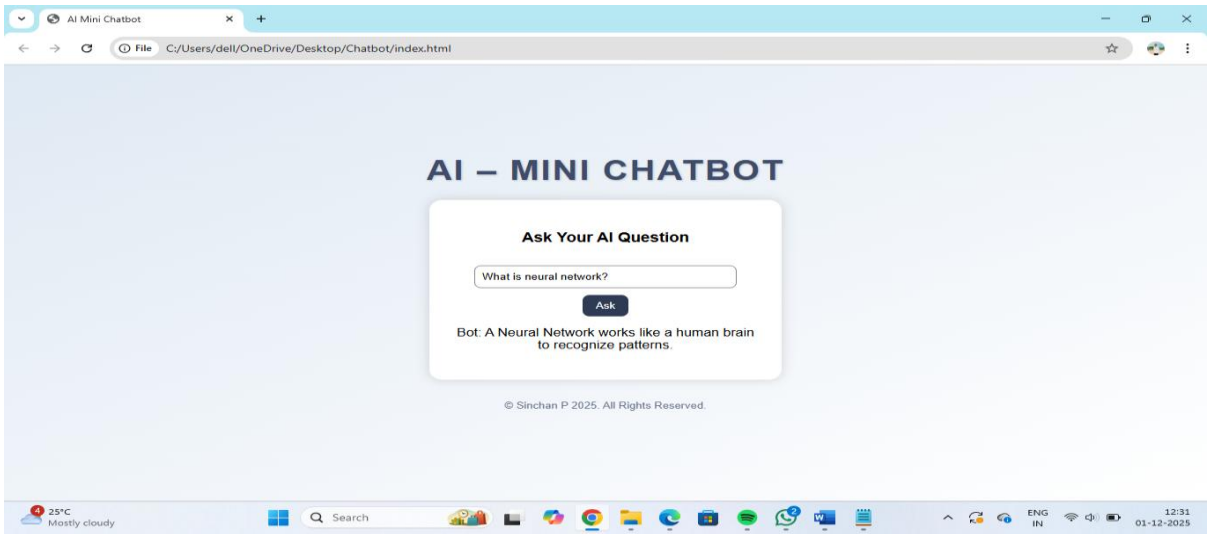
```
    </style>
</head>
<body>
    <h1>AI-MINI CHATBOT</h1>
  <div id="chatbox">
        <h3>Ask a Question</h3>
        <input type="text" id="userInput" placeholder="Type your question...">
        <button onclick="answer()">Ask</button>
        <p id="response"></p>  </div>
 <footer>© Sinchan P 2025 — All Rights Reserved</footer>
<script>
    function answer() {
        let q = document.getElementById("userInput").value.toLowerCase();
        let qa = {
            "what is ai": "AI allows machines to perform tasks that require human intelligence."
            "what is deep learning": "Deep Learning uses neural networks to large datasets."
            "what is a chatbot": "A chatbot is a computer program that interacts like a human."
        };
       let res = qa[q] || "Sorry, I know only basic AI questions.";
        document.getElementById("response").innerText = res;
    }
</script>
</body>
 </html>
```

CHAPTER 5

SNAPSHOTS







CHAPTER 6

CONCLUSION

The AI Mini Chatbot project successfully demonstrates the basic working of a rule-based chatbot system. Although modern AI uses advanced NLP and deep learning techniques, this project shows that even simple logic-based systems can simulate conversation in a controlled manner.

The chatbot responds to a set of predefined AI-related questions, showing how question–answer systems operate. It teaches students about user input handling, string comparison, UI design, and JavaScript logic.

This project also highlights the importance of:

- AI basics
- Human–computer interaction
- Rule-based systems
- User-friendly design
- Structured programming

Overall, the project is a useful starting point for understanding how chatbots work before moving to advanced technologies like Python NLP libraries or AI APIs.

CHAPTER 7

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